

Gen AI Academy

APAC Edition

Build in APAC. Build for the world.



Participant Details

- a. Participant Name: Sarvani Laxmi Prasanna
- a. Problem Statement: Ai for Better living and smart Communities

Brief about the idea

AI for better living and smart communities means using artificial intelligence to make daily life safer, easier, and more sustainable for people. In smart communities, AI analyzes data from traffic, energy grids, healthcare, and public services to make real-time decisions. For example, it can optimize traffic lights to reduce congestion, predict power outages to prevent them, monitor air quality for health alerts, and help hospitals manage patient care more efficiently. It also enables personalized services like smart waste management, water conservation, and faster emergency response. By automating routine tasks and providing useful insights, AI helps communities use resources better, improves quality of life, and creates cities that are more connected, inclusive, and resilient.

Your solution should be able to explain the following:

How did you approach the common problem statement and translate it into a working solution using Google Cloud technologies (e.g., BigQuery, ADK, Vertex AI, Gemini, etc.) while ensuring the hackathon's core requirements are met? If you selected the NVIDIA problem statement, also explain how NVIDIA technologies were integrated into your solution.

What real-world problem does your solution address, and what practical impact does it create for users, organizations, or communities??

What is the core architecture or workflow behind your solution, and how does it transform data into actionable insights or intelligent decisions?

1. Approach

BigQuery + Vertex AI + NVIDIA GPUs for prediction. ADK + Gemini via NIM for AI agent decisions.

2. Problem + Impact

Reduces loan defaults. Banks save money, customers get fair payment options.

3. Workflow

Data → Risk Score → AI Agent Acts → Feedback.

Opportunities

- How different is it from any of the other existing ideas?
- USP of the proposed solution

How it's different:

Existing tools are reactive + rule-based.

Ours predicts risk early, explains “why”, and acts automatically.

USP:

Proactive, Fair, and Autonomous Collections at Scale

Predicts → Explains → Acts. With Google Cloud + NVIDIA for 100% coverage, low cost.

List of features offered by the solution

1. Predictive Risk Scoring

Uses Vertex AI + NVIDIA GPUs to predict delinquency risk before it happens.

2. Explainable AI

Gives clear “reason codes” for each risk score so decisions are transparent.

3. Autonomous AI Agent

ADK + Gemini agent decides and executes the next best action automatically.

4. Personalized Customer Communication

Sends tailored SMS, email, or payment plan offers based on customer profile.

5. Real-time Monitoring

BigQuery pipeline monitors 100% of customers 24/7 instead of sample-based checks.

7. Feedback & Continuous Learning

Outcomes are logged back to BigQuery to retrain and improve the model.

Process flow diagram or Use-case diagram

[Customer Data]



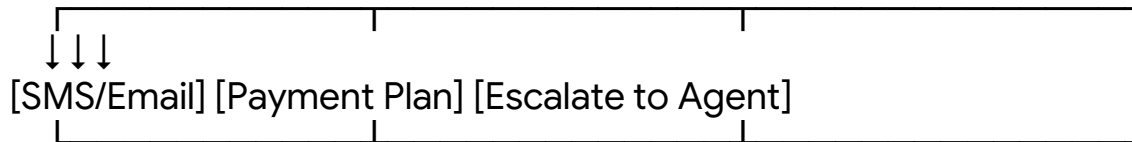
[BigQuery] → Data Ingestion & Storage



[Vertex AI + NVIDIA GPU] → Risk Prediction + Explainability



[ADK Agent + Gemini via NVIDIA NIM] → Decision Engine



[Feedback to BigQuery]

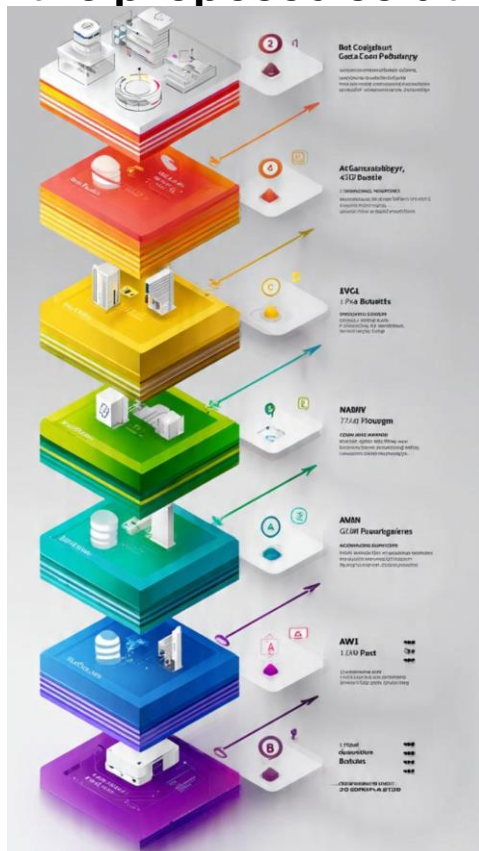


[Continuous Model Retraining]

Wireframes/Mock diagrams of the proposed solution



Architecture diagram of the proposed solution:



Technologies / Google / Nvidia Services used in the solution

(Why did you choose your specific AI stack and system design, and how does it support scalability and real-world deployment?)

1. Google Cloud

- BigQuery: Store + analyze 100% customer data in real-time
- Vertex AI: Train, deploy & monitor risk prediction models
- Cloud Functions + Pub/Sub: Trigger actions like SMS, emails instantly
- ADK: Build autonomous AI agents for decision making

2. NVIDIA

- GPU Acceleration on Vertex AI: 10x faster model training + inference
- NVIDIA NIM: Low-latency Gemini LLM inference for the AI agent

Snapshots of the prototype

Snapshot 1: Admin Risk Dashboard

Shows real-time risk overview

- Total customers, At-Risk count, Predicted recovery
- Risk heatmap + Top 3 risk reasons
- “High Priority Queue” table with Risk Score, Reason, and [Take Action] buttons

Snapshot 2: AI Agent Workbench

Shows how the autonomous agent decides

- Customer profile: Risk 89%, Reason: “Salary delayed 2 months”
- Gemini drafts message: “Hi Rahul, need a 7-day extension?”
- Action buttons: [Send SMS] [Offer Payment Plan][Escalate]

Snapshot 3: Customer Self-Serve Portal

What the customer sees on mobile/web

- Due amount + due date
- AI suggested options: “Pay in 3 EMIs” or “Get 7-day Extension”
- Note: “Personalized to help you avoid late fees”

Google Cloud 

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Thank you

